

Algebra II with Trigonometry

Chapter 7.2

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4582337_alg-2trig-h-chp-7-2-note-docx.pdf](#)

Based on your PDF (covering trigonometric addition/subtraction formulas, expressions with sine and cosine, inverse trig, and exact values), here are **10 mental math questions** that test the core concepts and can be solved without pencil and paper:

1.

What is $\sin(0^\circ + 90^\circ)$?

2.

Find $\cos(180^\circ - 0^\circ)$.

3.

If $\tan 45^\circ = 1$, what is $\tan(45^\circ + 45^\circ)$ using the addition formula?

4.

If $\sin \alpha = 1$, what is $\cos(90^\circ - \alpha)$?

5.

True or False: $\cos(180^\circ + x) = -\cos x$.

6.

What is the exact value of $\cos(0)$?

7.

If $A = 3$, $B = 4$, what is $\sqrt{A^2 + B^2}$?

8.

What is the value of $\tan^{-1}(1)$ in degrees?

9.

If $\sin x = 0$, what is $\cos(x)$ when x is in the first quadrant?

10.

What is $\sin(30^\circ + 60^\circ)$ using the addition formula and known values?

Let me know if you'd like answers or explanations for any/all!